$$\frac{|\Psi(\epsilon)\rangle - |\Psi(0)\rangle}{\epsilon} = -i\mathbf{H}|\Psi(0)\rangle.$$

If you remember your calculus (see *Volume I* for a quick review), you'll recognize that the left-hand side of this equation looks exactly like the definition of a derivative. If we take the limit as $\varepsilon \to 0$, it becomes the time derivative of the state-vector:

$$\frac{\partial|\Psi\rangle}{\partial t} = -i\mathbf{H}|\Psi\rangle. \qquad (4.9)$$

We originally set things up so that the time variable was zero, but there was nothing special about $t = 0$. Had we chosen another time and done the same thing, we would have gotten exactly the same result, namely, Eq. 4.9. This equation tells us how the state-vector changes: if we know the state-vector at one instant, the equation tells us what it will be at the next. Eq. 4.9 is important enough to have a name. It is called the *generalized Schrödinger equation*, or more commonly, the *time-dependent Schrödinger equation*. If we know the Hamiltonian, it tells us how the state of an undisturbed system evolves with time. Art likes to call this state-vector *Schrödinger's Ket*. He even wanted to render the Greek symbol with little whiskers,[1] but I had to draw the line somewhere.

4.6 WHAT EVER HAPPENED TO ħ?

I'm sure you have all heard of Planck's constant. Planck himself called it *h* and gave it a value of about 6.6×10^{-34} *kg* m^2 */s.*

Later generations redefined it, dividing by a factor of 2π and calling the result *ħ*:

$$\hbar = \frac{h}{2\pi} = 1.054571726\cdots\times10^{-34}\ kg\ m^2/s.$$